

KidCode 3

WEEK 1: LIBRARIES, NLP & SETUP

Day 1: Welcome Back & Skills Check-Up

Reviewing Years 1 and 2 with "Python Olympics 2.0": functions, classes, files, Counter, and comprehensions. Revisiting Bug Safari: why `.split()` fails on "Dr. Lee lives in the U.S.A." and "don't".

Day 2: What Is a Library, Really?

Library vs. package vs. module vs. API. The anatomy of an `import`, and how libraries are built from the same functions and classes students wrote in Year 2. A "Standing on Shoulders" scavenger hunt on PyPI.

Day 3: Installing Libraries Safely

Using `pip install`, `pip list`, and `pip show`, creating a virtual environment with `python -m venv`, and saving `requirements.txt`. Reading install errors. Lab: "Project Sandbox", a clean environment for the year.

Day 4: What Is NLP?

Natural Language Processing in daily life: autocorrect, voice assistants, translation, and search. Why language is hard for computers: ambiguity ("I saw her duck"), slang, and punctuation.

Day 5: Meet spaCy

Installing `spacy` and the `en_core_web_sm` model, loading it with `spacy.load()`, and passing text to the `nlp` object to get a Doc made of Token objects. Lab: "Hello, spaCy".

WEEK 2: TOKENIZATION

Day 6: Tokenization: Beyond split()

What a token is. Comparing `text.split()` with spaCy's tokens on punctuation, contractions (don't becomes do and n't), URLs, and numbers. A tokenizer-only pipeline with `spacy.blank("en")`.

Day 7: Token Attributes

`token.text`, `token.is_alpha`, `token.is_punct`, `token.is_digit`, `token.like_num`, and `token.like_url`. Indexing and slicing a Doc with `doc[0]` and `doc[1:4]`. Lab: "Token Sorter".

Day 8: Stop Words & Text Cleaning

`token.is_stop`, `token.lower_`, and filtering tokens with list comprehensions. Building a cleaning recipe that lowercases text and drops punctuation and stop words. Practice: "Clean-Up Crew".

Day 9: Lemmas & Normalization

`token.lemma_`: how "running", "ran", and "runs" all become "run". Why the trained model is needed for lemmas, and how normalization makes counting fairer. Practice: "Word Family Finder".

Day 10: Lab: Tokenizer Showdown

Project 1. Testing `.split()`, Year 2's cleaning code, and spaCy on twenty tricky sentences, scoring each on contractions, abbreviations, and punctuation, and charting the results.

WEEK 3: SENTENCES & GRAMMAR

Day 11: Sentences & Spans

Splitting text with `doc.sents` and working with Span objects such as `doc[2:5]`, including `.text`, `.start`, and `.end`. Comparing spaCy's results with Year 2's `.split(" ")`.

Days 12-13: Lab: Keyword Detective

Project 2. Tokenizing a chapter, dropping stop words and punctuation, lemmatizing, counting with Counter, and charting the top keywords. Comparing with Year 2's Book Detective on the same text.

Day 14: Parts of Speech

Nouns, verbs, adjectives, and adverbs, and how spaCy labels them with `token.pos_` and `token.tag_`. Decoding labels with `spacy.explain()`. Unplugged game: "Human Sentence".

Day 15: Filtering by Part of Speech

Using comprehensions to pull out NOUN, VERB, and ADJ tokens. Finding the most common verbs in a fairy tale and exploring where the model is tricked ("Time flies like an arrow").

WEEK 4: GRAMMAR DETECTIVES

Day 16: Lab: Auto Mad Libs I

Project 3. Revisiting Year 1's Mad Libs: spaCy finds the nouns, verbs, and adjectives in a story and blanks out a random sample. Planning the program with a flowchart first.

Day 17: Lab: Auto Mad Libs II

Asking the player for words of the right type, then rebuilding the story with `token.text_with_ws` so spacing and punctuation stay correct. Swapping finished stories with classmates.

Day 18: Dependency Parsing: Who Did What?

`token.dep_`, `token.head`, and `token.children`, and finding subjects and objects. Drawing sentence diagrams with `displacy.render()`. Practice: "Sentence Skeletons".

Day 19: The spaCy Pipeline

The pipeline as an assembly line: `nlp.pipe_names` and what each component adds. Processing many texts with `nlp.pipe()` and turning off unused components for speed.

Day 20: Checkpoint: spaCy Cheat Sheet

Review games covering Weeks 1 to 4, a checkpoint quiz, and a personal spaCy cheat sheet. Pair debugging with error messages students have not seen before.

WEEK 5: NAMED ENTITY RECOGNITION

Day 21: What Are Named Entities?

Real-world names and facts hiding in text. Using `doc.ents`, `ent.text`, `ent.label_`, and `ent.start_char`, and decoding PERSON, ORG, GPE, DATE, and MONEY with `spacy.explain()`.

Day 22: Seeing Entities with displaCy

Highlighting entities with `displacy.render(doc, style="ent")` in a notebook or `displacy.serve()` in the browser, and saving results as HTML. Lab: "Entity Gallery" of student paragraphs.

Day 23: When the Model Is Wrong

Why models guess: ambiguous names like "Washington", "Apple", and "Jordan", names from many cultures, and text without capital letters. Hand-labeling 20 sentences for an "NER Report Card".

Day 24: Lab: Character Tracker I

Project 4. Loading a public-domain story and running one chapter at a time through `nlp`. Extracting PERSON and GPE entities and counting them with `Counter`.

Day 25: Lab: Character Tracker II

Merging name variants like "Dorothy" and "Dorothy Gale", tracking which characters appear in each chapter, and printing a "Who's Who" report. Listing what the tool gets wrong.

WEEK 6: RULES & CUSTOM ENTITIES

Day 26: Lab: Privacy Shield I

Project 5. Planning a redaction tool: which entity types (PERSON, GPE, ORG, DATE) to hide, what placeholders like [PERSON] look like, and how to test on sample letters.

Day 27: Lab: Privacy Shield II

Replacing entities while keeping the text readable, using `token.ent_iob_` and `token.text_with_ws` so multi-word names are replaced once. Testing for names the model missed.

Day 28: Rule-Based Matching

Building patterns from token attributes such as `{"LOWER": "hello"}`, `{"IS_PUNCT": True}`, and `{"POS": "NOUN"}`. Using `Matcher`, `matcher.add()`, and `matches as (match_id, start, end)`.

Day 29: Teaching spaCy New Entities

Adding an entity ruler with `nlp.add_pipe("entity_ruler", before="ner")`, writing patterns for custom labels like FOOD or PET, and seeing how rules and the model work together.

Day 30: Lab: Fantasy Map Finder

Recognizing invented places, heroes, and creatures with entity ruler patterns, then highlighting them with `displaCy`. Testing what happens when a rule and the model disagree.

WEEK 7: CHATBOTS & RESPONSIBLE NLP

Day 31: Lab: KidBot Brain I

Project 6. Designing a rule-based chatbot: listing intents (greeting, joke, weather), writing `Matcher` patterns for each, and sketching the conversation loop.

Day 32: Lab: KidBot Brain II

Using entities to pull out places and dates so "What's the weather in Paris tomorrow?" gets a pretend forecast. Adding a friendly fallback reply and testing with classmates.

Day 33: How Models Learn

How `spaCy`'s model learned from labeled examples, why training data shapes what a model gets right, and how mistakes and bias appear. Students annotate 20 sentences and compare with the model.

Day 34: Responsible NLP: Bias & Privacy

Fairness, privacy, and knowing when not to use NLP. Class roundtable: "Should a school use NLP to read student essays?" Writing a "Limitations" card for a personal project.

Day 35: Capstone Pitch & Plan

Choosing a project: "Story Explorer", "KidBot Pro", "Privacy Shield Pro", "Time Traveler's News Desk", or an approved original idea. A one-page plan with goals, test texts, and a feature checklist.

WEEK 8: CAPSTONE & SHOWCASE

Day 36: Capstone Build I

Project 7. Loading text from a file, running the `spaCy` pipeline, and organizing code into functions or classes. Daily stand-up check-ins: what I did, what is next, and what is blocking me.

Day 37: Capstone Build II

Adding features: `Counter` charts, a `displaCy` HTML export, `Matcher` or entity ruler rules, and `try/except` for missing files. Writing a comment header that explains how to run the program.

Day 38: Beta Test & NER Audit

Swapping projects so testers can try to break them with tricky texts. Running an NER audit on each tool and logging every wrong guess in a bug list.

Day 39: Polish & Demo Rehearsal

Fixing bugs, cleaning up code, and writing "How It Works" and "Limitations" cards. Rehearsing a three-minute demo with a partner.

Day 40: Showcase Day & What's Next

Presenting projects to classmates, families, and guests, and celebrating three years of KidCode with certificates. Next steps: machine learning with `scikit-learn`, web apps with `Flask`, and games with `pygame`.